Al buio non si trova

Biostatistics in the 21st century

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Available from: https://github.com/maxbiostat/presentations/

FGV EMAp

Plan for today

Problem I: using historical data to the fullest

Optimal Bayesian dynamic borrowing of information

Problem II: dealing with huge complex data

MCMC in tree space: a journey through a strange land

Where the magic happens

Academic life at FGV EMAp

Problem I: efficiently utilising available information

Loads of historical data: how to build informative priors?

Let $\boldsymbol{y_0} = (y_{01}, \ldots, y_{0n_0})$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$ be **historical** and **current** data, respectively.

Question: how do I build a prior that

- Uses information in $\boldsymbol{y_0}$ efficiently but also
- Does not lead to borrowing too much information when the data sets are not compatible?

Applications: clinical trials, quality control, policy-making.

I got the power

Normalised power prior (NPP)[1]

$$\tilde{\pi}(\theta, a_0 \mid \boldsymbol{y}_0) \propto \frac{L_0(\boldsymbol{y}_0 \mid \theta)^{a_0}\pi_0(\theta \mid \phi)\pi_A(a_0 \mid \xi)}{c(a_0;\phi)}$$

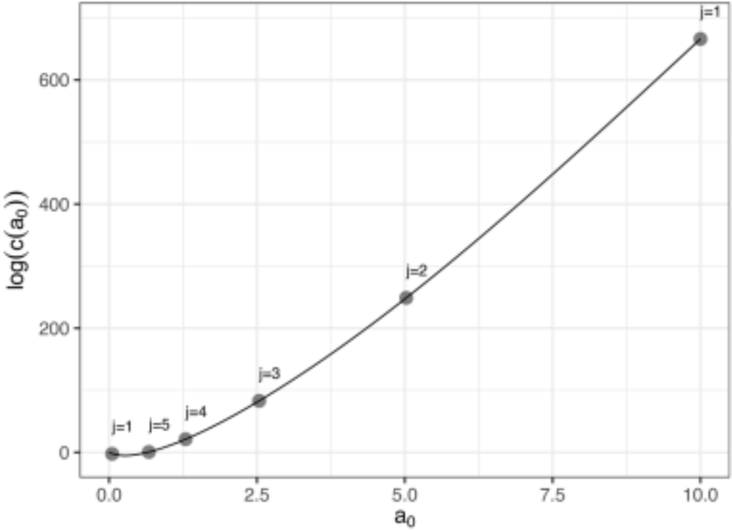
- How pick π_A such that prediction error (say) is minimised?[2]
- How to **efficiently** compute

$$c(a_0;\phi) = \int_\Theta L_0(\boldsymbol{y}_0 \mid t)^{a_0}\pi_0(t \mid \phi)\, d\mu(t)$$

by leveraging its special properties as function of a_0?

[1] Carvalho & Ibrahim (2021) *Statistics in Medicine*.
[2] Shen et al. (2024) *Statistica Sinica*.

Approximating the normalising constant



What we are doing right now

Principled Bayesian analysis under the NPP[3]

Calibrate the initial prior π_0 and pick the prior on the discounting parameter a_0 *via* prior predictive checking.

Exact MCMC for the NPP[4]

The approximate version comes with no guarantees

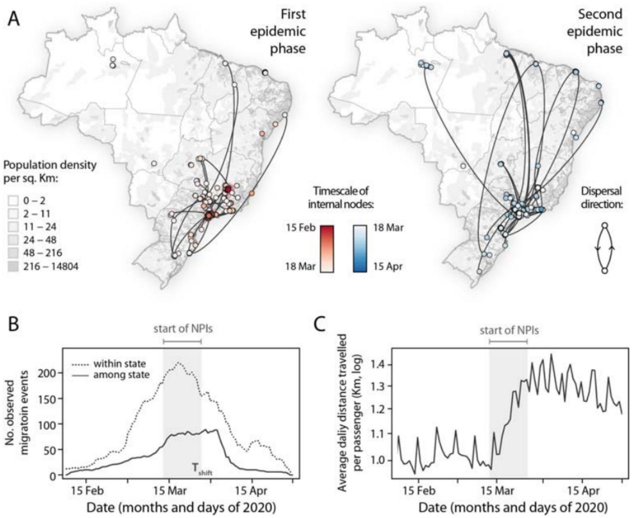
Variational approximations for large data/models[5]

Sometimes approximate is all you have, but need provable performance

[3] Ezequiel Braga (MSc)
[4] Eduardo Adame (MSc)
[5] Yure Oliveira (IC)

Problem II: dealing with huge complex data

Where did this virus come from?[6]



[6] Cândido et al. (2020) *Science*.

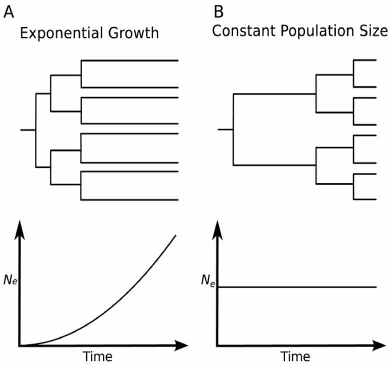
Motivation

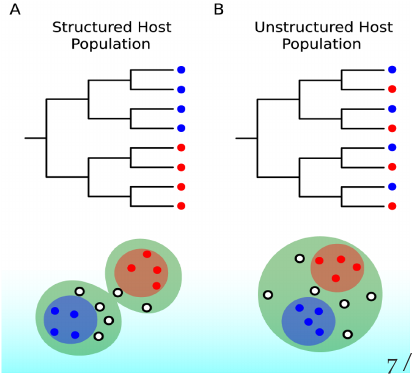
Phylodynamics of fast-evolving viruses

Inferring spatial and temporal dynamics from genomic data:

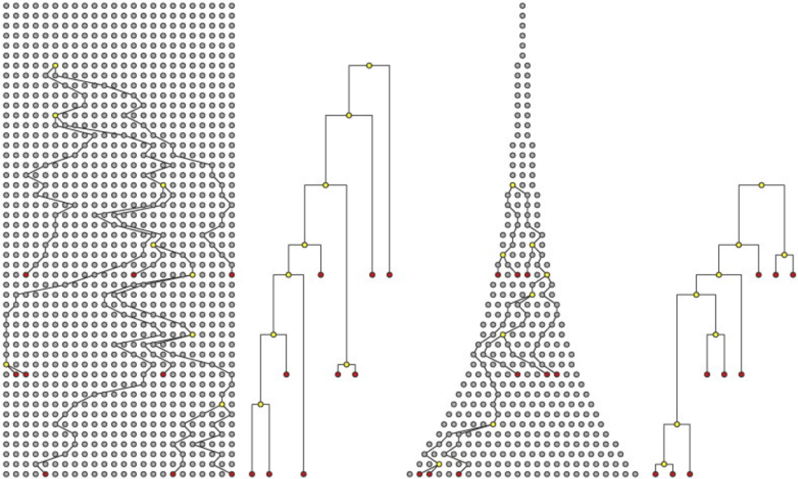
Phylogenies*!

* plus complicated models





Trees and the coalescent



Central object: time-calibrated trees

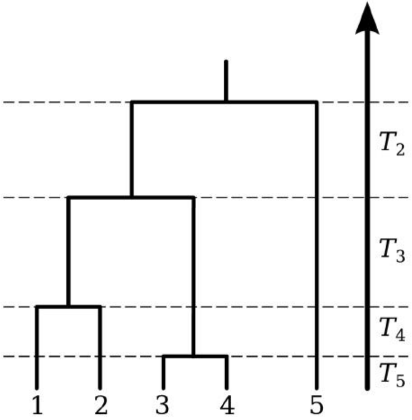


Figure: Figure 4 from Volz et al. (2013).

Let T_n denote the time for n lineages to *coalesce*, i.e., merge into one ancestral lineage, in a population of size N_e. Then:

$$Pr(T_n = t) = \lambda_n e^{-\lambda_n t}$$

$$\lambda_n = \binom{n}{2} \frac{1}{N_e} = \binom{n}{2} \frac{1}{N_e \tau}$$

where N_e is the effective population size and τ is the generation time. Let T_{mrca} denote the age of the most recent common ancestor:

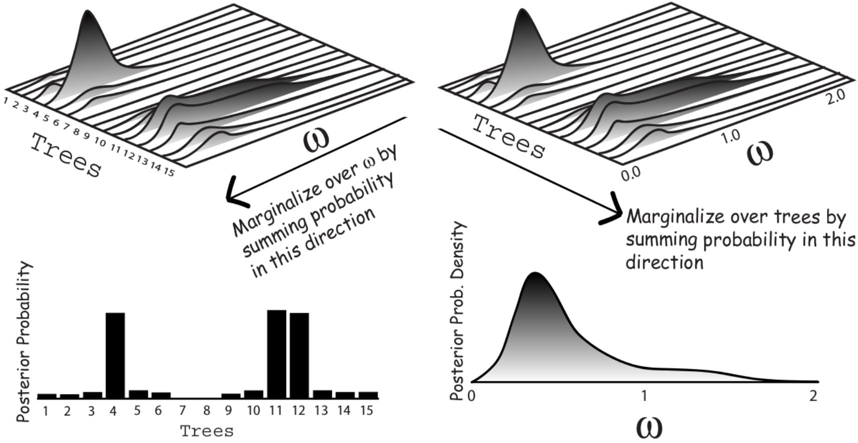
$$\begin{aligned}
\mathbb{E}[T_{\text{mrca}}] &= \mathbb{E}[T_n] + \mathbb{E}[T_{n-1}] + \ldots + \mathbb{E}[T_2] \\
&= 1/\lambda_n + 1/\lambda_{n-1} + \ldots + 1/\lambda_2 \\
&= 2N_e \left(1 - \frac{1}{n}\right)
\end{aligned}$$

Target

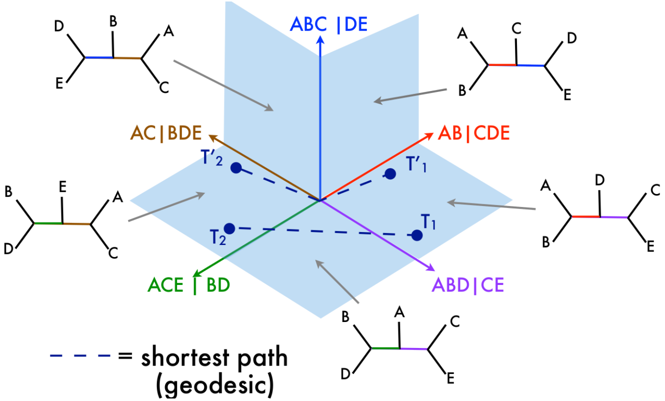
$$p(t, \boldsymbol{b}, \boldsymbol{\omega} | D) = \frac{f(D|t, \boldsymbol{b}, \boldsymbol{\omega})\pi(t, \boldsymbol{b}, \boldsymbol{\omega})}{\sum_{t_i \in \boldsymbol{T}_n} \int_{\boldsymbol{B}} \int_{\boldsymbol{\Omega}} f(D|t_i, \boldsymbol{b}_i, \boldsymbol{\omega})\pi(t_i, \boldsymbol{b}_i, \boldsymbol{\omega}) d\boldsymbol{\omega} d\boldsymbol{b}_i} \quad (1)$$

- D: observed sequence (DNA) data;
- $\boldsymbol{T}_n$: set of all binary ranked trees;
- $\boldsymbol{b}_k$: set of branch lengths of $t_k \in \boldsymbol{T}_n$ ($\mathbb{R}_+^{2n-2}$, kind of) ;
- $\boldsymbol{\omega}$: set of parameters of interest such as substitution model parameters, migration rates, heritability coefficients, etc.

The end product



Traversing cubic complexes efficiently[7]



Applications: Molecular Epidemiology, Evolutionary Biology.

[7] https://youtu.be/h9bWRQ6aeKA

General MH setup.

Let $\tau = (t, \boldsymbol{b})$ denote a tree with topology t and branch lengths $\boldsymbol{b}$. For two trees τ and τ', denote the transition kernel by $q_\gamma(\tau|\tau') := \Pr(\tau' \to \tau|\gamma)$.

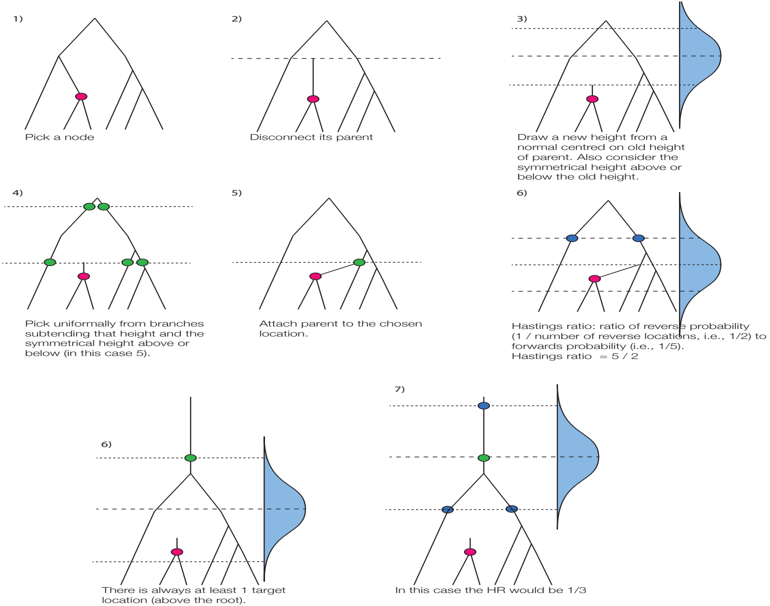
Accepting with probability

$$A_\gamma(\tau|\tau') = \min\left(1, \frac{p(\tau', \boldsymbol{\omega}|D)q_\gamma(\tau|\tau')}{p(\tau, \boldsymbol{\omega}|D)q_\gamma(\tau'|\tau)}\right)$$

leads to the desired target.

Note: Here $\gamma > 0$ is a so-called tuning parameter.

STL – illustration



STL – ergodicity

Carvalho (2019), Chapter 2.

Lemma

Assume strictly positive branch lengths. Then SubTreeLeap induces an irreducible Markov chain on $\boldsymbol{T}_n$.

Sketch: Starting at $x \in \boldsymbol{T}_n$, notice there exists $\delta_y^\star > 0$ such that $P\left(x \to y \mid \delta_y^\star\right) > 0$ for any tree $y \in \boldsymbol{T}_n$ in the SPR neighbourhood of x.

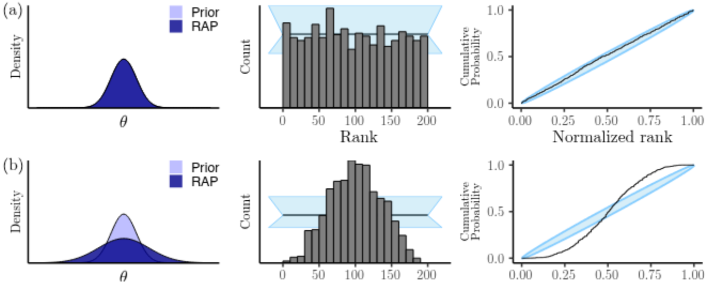
Theorem

Assume the target satisfies $p(A) > 0$ for all $A \subset \boldsymbol{\Psi}$. Then, SubTreeLeap induces an ergodic Markov chain on $\boldsymbol{\Psi}$.

Sketch: Employ the lemma to get to the case where $d_{\text{SPR}}(x, y) = 0$ and then establish Harris recurrence.

How do you tell if it worked, though?

- **Super important** to check consider actual tree metrics when checking convergence (convergence on ω is not informative)[8].
- **Validating** the methodology using rigorous modern techniques such as Simulation-based Calibration (SBC)[9]



[8] Brusselmans et al (2024) *Virus Evolution*

[9] Mendes et al. (2025) *Systematic Biology*

Open problems in MCMC for phylogenies

Open problems:

- Theoretical aspects of phylogenetic MCMC[10];
- How can we construct more efficient proposals? How to exploit structure? **Geometry!**
- How to quantify exploration of the target? (Custom) **Tools!**
- Optimal scaling: what's the **optimal** acceptance probability?
- New, statistically-motivated metrics and measures of centrality (e.g. HIPSTR[11].)

[10] See Semper & Palacios (2022) and Alves, Saporito & Carvalho (2022) for recent theoretical attempts and Gao et al. (2026) for an empirical investigation.
[11] Baele et al (2025) *Bioinformatics*.

Onde a mágica acontece – FGV EMAp

A ESCOLA

Excelência na aplicação da matemática e da computação

Com método de ensino participativo e professores altamente qualificados, a FGV EMAp prepara profissionais para desenvolverem matemática e modelos computacionais de forma inovadora, adaptados aos desafios da era da informação.

Aqui, os alunos saem com uma base sólida para atuar em setores estratégicos de organizações públicas e privadas, qualificando-os, também, para a realização de pesquisas acadêmicas e projetos de consultoria.

Conheça a escola

FGV EMAp website

Pós na EMAp: seleção

- Mestrado e Doutorado em **Matemática Aplicada e Ciência de Dados**;
- Edital aberto até 30/9;
- Seleção: análise de currículo, e se aprovado, entrevista;
- **Bolsa de mestrado: R$ 3.000,00;**
- **Bolsa de doutorado: R$ 4.000,00;**
- Possibilidade de doutorado direto;
- Bolsas de projetos e monitoria.

Pós na EMAp: estrutura

Mestrado

- 7 disciplinas: 4 centrais e 3 eletivas
- 2 disciplinas de seminários

Doutorado

- 7 disciplinas: 3 centrais e 4 eletivas
- 4 disciplinas de seminários
- Exame de Qualificação I, II, III e Tese

Disciplinas

Centrais do Doutorado (e Mestrado)

Medida, Integração e Probabilidade	Otimização
Teoria Qualitativa de EDO	Controle Ótimo
Análise Numérica e Simulação	Processos Estocásticos
Análise Funcional: Fundamentos	Equações Diferenciais Estocásticas: Teoria e Simulação
Análise Funcional: Operadores Lineares	Finanças Quantitativas
Ciência de Redes	Métodos Quantitativos em Risco
Equações Diferenciais Parciais e Aplicações	Biologia Matemática
Visualização para Aprendizado de Máquinas	Causalidade
Aprendizado de Máquinas	Epidemiologia Matemática
Modelos Generativos	Inferência Estatística
Redes Neurais e Aprendizado Profundo	Estatística Computacional
Aprendizado por Reforço	Estatística Bayesiana

Nossos egressos

Lucas Moschen
Imperial College London (antes, Sorbonne)

Lucas Resck
Cambridge University

Vitória Guardieiro
University of Pennsylvania

Take home

A light in the dark

Scalable methods with provable guarantees

Computational methods are key

Learn to program and learn Computational Statistics[12]

We've got loads to do!

Biomedical statistics is where most of the cool data and problems **with actual impact** are.

Come work with us!

Come to the School of Applied Mathematics (FGV EMAp)

[12] Here's a place to start: https://github.com/maxbiostat/Computational_Statistics

THE
END